SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I _PREETHI R/192511352__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

SET 1

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Cloud storage VS SAN - Cost, Transfer Time and Security

a) Given

San capacity = 200 TB

SAN purchase cost = ₹24,00,000

Annual operating cost = ₹ 3,00,000

Cloud Storage cost = ₹0.50 per GB per month

Monthly egress = 2TB at ₹5 per GB

Data grows from 100TB to 180 TB over five years

option A: SAN cost

Total Cost = purchase cost + operating cost

= 24,00,000 + (5 × 3,00,000)

= 39,00,000

option B: Cloud Storage cost

Cost per TB per year:

0.50 × 1000 × 12 = ₹6,000

Total Storage over five years:

100 + 120 + 140 + 160 + 180 = 700 TB-years

→ Storage cost: 700 × 6,000 = ₹42,00,000

→ Egress cost: 2000 × 5 × 60 = ₹6,00,000

→ Total Cloud Cost : 42,00,000 + 6,00,000 = ₹ 48,00,000

→ SAN Cost Saving : 48,00,000 - 39,00,000 = ₹ 9,00,000

SAN utilisation :

Year 1 : $\frac{100}{200} \times 100 = 50\%$.

Year 5 : $\frac{180}{200} \times 100 = 90\%$.

SAN is ₹ 9,00,000 Cheaper over five years. the data reaches 200 TB at the start of year 6. So the college will need to expand its SAN capacity

b) Given

Usable Speed is 80%

Convert 100 TB into bits : $100 \times 10^6 \times 8 = 8 \times 10^4$ bits

Internet transfer:

Effective Speed : $1 \times 0.8 = 0.8$ Gbps

Transfer time : $\frac{8 \times 10^4}{0.8 \times 10^9} = 10,00,000$ Sec

$\frac{10,00,000}{86,400} = 11.6$ days

SAN transfer:

Effective Speed : $16 \times 0.8 = 12.8$ Gbps

Transfer time : $\frac{8 \times 10^{14}}{12.8 \times 10^9} = 62,500$ Sec

$\frac{62,500}{3600} = 17.4$ hours

The SAN fabric is 16 time faster than the internet link. The college should consider a faster internet connection or an offline data-transfer service for the initial cloud upload.

c) Compare SAN architecture with cloud storage in acess method.

Feature	SAN	cloud Storage
Access method	Serve connect through host bus adapters and fibre channel Switches to shared stored arays	Accessed over the internet through web interface and APIs, usually as object or files.
Scalability	capacity is increased by purchasing additional hardware	Capacity grows on demand
pufarmance	offers high speed and low latency	Depends on internet speed and network latency
Reliability	Uses RAID, redundant controller and local backups	uses data replication across redundant infrastructure.

d) Security and availability issue

SAN issue:

→ Hardware or site failure may cause data loss

→ The college must manage Security, update and backups.

Cloud Storage issues:

→ privacy risks and Stolen Crendentials may expose data.

→ Internet outage and egress charge may cause problem.

e) Recommended Storage Solution

A hybrid Storage Solution is recommended

The SAN costs £37,00,000 over five years, saving £900,000 compared with cloud Storage. It also provides faster access to frequently used recording

The College Should Store current recording on the SAN and keep archieved recording backup and disaster recovery copies in encrypted cloud Storage. This reduces SAN usage and delays expansion in year 6.

2. Web API Access, Network capacity and collaboration in the Cloud.

a) Given:

Request = 4,000 in 5 Sec

Data per request = 25 KB

Link capacity = 100 Mbps

Gateway capacity = 1,000 request/s

Request rate = 4,000 ÷ 5 = 800 request/s

Bandwidth required = 800 x 25 KB = 20 MB/s

= 20 x 8 = 160 Mbps

Gateway utilisation: (800 ÷ 1,000) x 100

= 80%

Number of Support users

per-user bandwidth = (25 x 8) ÷ 5

= 40 kbps

Users Supported = 1,00,000 ÷ 40

= 2,500 users

The gateway operates at 80% utilisation, but the network requires 160 Mbps while only 100 Mbps is available. The network is the bottleneck.

b) Evaluate caching and compression

Caching stores frequently requested data so that it can be reused without repeatedly contacting the server. Compression reduces the size of data transfered over the network.

Method	Calculation	Bandwidth Required
Caching only	800 x 40% x 25 KB x 8	64 Mbps
Compression only	800 x 12.5 KB x 8	80 Mbps
Both method	480 x 12.5 KB x 8	48 Mbps

This users 48% of the available link capacity and provide sufficient headroom. caching and compression reduce network load and improve response time.

c) Compare thin and thick client

Thin client (Browser)	Thick client
* No installation required	Require installation
* Easy central updates	update on each device
* Low maintanance cost	Higher maintenance cost
* Require internet ; limited offline use	Support local processing and offline work.
* Work across different devices	May depend on the operating system.

d) Security issues and controls:

1. Data interception during transmission
2. Stolen API keys and unauthorized access.
3. Injection attacks, Cross-Site Scripting and Session hijacking
4. API abuse and data leakage through Shared links.

Controls: Use HTTPS/TLS, OAuth 2.0, multi-factor authentication

e) Recommend a Suitable Architetwe.

A browser-based application using REST/JSON Web APIs is recommended. It should use HTTP/HTTPS

and OAuth 2.0 for secure communication.

Real time collaboration can be supported using WebSocket or push notification, with a central shared document and version history instead of exchanging files through email.

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3. Cloud computing and Big data analytics

a) A Smart City can use a three-layer cloud architecture

1. Data collection layer: Cameras, Sensors and citizen application collect data.

2. Cloud infrastructure layer: Virtual machines, Storage and processing resources store and process the data.

3. Application layer: provides traffic monitoring, congestion analysis and citizen services.

REST APIs using HTTPS and JSON enable communication between devices and cloud application. Cloud Storage scale on demand, while load balancing distributes request across servers. Edge processing, compressing and caching reduces network bandwidth requirement.

Challenges and Solution:

* Security: Unauthorized access and data leakage can be prevented using encryption, multi-factor authentication

role-based access control and audit logs.

* Availability: Cloud outages and network failures can be managed using redundant connection, replicated Storage, regular backups and disaster recovery.

* Scalability: Increasing data volume can be handled using Scalable cloud resource and distributed Storage.

Recommendation: use Scalable cloud infrastructure with Serverless APIs, load balancing and edge processing for reliable Smart-city Services.

b)

1. Daily data volume Calculation

Video data:

= 2000 MB/s × 86,400 Sec

= 172,800,000 MB

= 172,800 GB/day

Sensor data:

Events per Second = 50,000 ÷ 2

= 25,000

Data per Second = 25,000 × 100 bytes

= 2.5 MB/s

Daily Sensor data = 2.5 × 86,400

= 216 GB/day

Total daily data

Video = 173,800 GB

Sensors = 216 GB

Citizen records = 20 GB

Total = 173,036 GB
≈ 173.04 TB/day

Yearly Storage requirement:

173.036 × 365 = 63,158.14 TB
≈ 63.2 PB/Year.

A 500 TB Storage System last approximately 500 ÷ 173.04 = 2.9 days. making it unsuitable for long term.

2. Data Classification
   - → Structured : citizen records stored in tables
   - → Semi Structured : Sensor message in JSON formal
   - → Unstructured : Traffic camera video.

3. Distributed Storage and analysis
   - * distributed file system.
   - * Stream processing
   - * Batch analytics
   - * Edge analytics
   - * privacy protection

4. Distributed file System

a) Given:

Number of node = 30

Storage per node = 12 TB

Usable capacity = 80%

Replication factor = 3

Daily data = 4TB

Raw capacity : $30 \times 12 = 360$ TB

Usable capacity : $360 \times 0.8 = 288$ TB

Unique data capacity : $288 \div 3 = 96$ TB

Retention : $96 \div 4 = 24$ days

Therefore, the System can retain 24 days of data.

b) Given:

Data Size = 60 TB = 60×10^6 MB

processing rate per Server = 200 MB/s

Number of nodes = 30

Single Server:

Time = $(60 \times 10^6) \div 200$

= 300,000 Sec

= 83.3 hours

Using 30 nodes:

Total rate = 30 × 200 = 6,000 MB/s

Time = $(60 \times 10^6) \div 6{,}000$

= 10,000 second

≈ 2.8 hours

Speed up:

Speed up = 83.3 ÷ 2.78

≈ 30 times

Thus parallel processing significantly reduces processing time, assuming data is evenly distributed and overhead is negligible.

c) Fault Tolerance and Recovery Time

When a node fails, the system copies its block from surviving replicas to other nodes

Recovery time = $(9.6 \times 10^6) \div 1{,}000$

= 9600 seconds

= 2.7 hours.

With three replicas on different node failure, provide the remaining replicas are available.

4)

Distributed file	Relational database
* Scales by adding nodes	Scales by upgrading hardware
* Support different data formats	uses a fixed schema.
* Enables parallel processing	Suitable for SQL queries and transactions
* Suitable for large logs and analytics	Suitable for structured data.

c) Calculate Nodes for 90 Day

Unique data = 4 × 90
= 360 TB

with 3 replicas = 360 × 3
= 1,080 TB

Raw capacity = 1,080 ÷ 0.8
= 1,350 TB

Nodes = 1,350 ÷ 12 = 112.5
≈ 113 nodes

Additional node = 113 - 30
= 83 node

Raw capacity = 1,350 ÷ 3
= 450 TB

Nodes = 450 ÷ 12 = 37.5 nodes

Recommendation

→ Compress logs and archive older data.

→ Use two replis for cold data where suitable

→ Apply load balancing and rack-aware placement.

5. Customer churn prediction

a)

Total Subscribers = 20,000,000

Actual churn rate = 2%.

Actual churn rate = 2% × 20,000,000

= 4,00,000

True positive = 2,80,000

False positive = 10,00,000 - 2,80,000

= 7,20,000

Missed churners = 4,00,000 - 2,80,000

= 1,20,000

b) Calculate recall, precision

Recall:

2,80,000 ÷ 4,00,000 ×100

= 70%.

Precision:

$= 2,80,000 \div 10,00,000 \times 100$

$= 28\%$

Lift:

$= 28\% \div 2\%$

$= 14$ times

The model identifies 70% of actual churners and perform 14 times better than random targeting.

(1) Campaign cost $= 10,00,000 \times ₹150$

$= ₹15$ crore

Subscribers retained $= 40\% \times 2,80,000$

$= 1,12,000$

Retained Subscribe Value $= 1,12,000 \times ₹6,000$

$= ₹67.2$ crore.

Net benefit $= ₹67.2$ crore $- ₹15$ crore

$= ₹52.2$ crore per month

Cost per retained Subscriber $= ₹15,00,000 \div 1,12,000$

$\approx ₹1,339$.

d)

→ precision = 44% ; recall = 55%

→ Campaign cost = 5,00,000 × ₹150

= ₹7.5 crore.

→ Retained Subscribers = 40% × 2,20,000

= 88,000

→ Retained value = 88,000 × ₹6,000

= ₹52.8 crore

→ Net benefit = ₹52.8 - ₹7.5 crore

= ₹45.3 crore

→ Cost per retained Subscriber = ₹7.5 crore ÷ 88,000

= ₹852

e) Security measures:

* Encrypt and anonymise Sensitive data
* Apply role-based access control.
* Maintenance audit trails and follow privacy requirements.